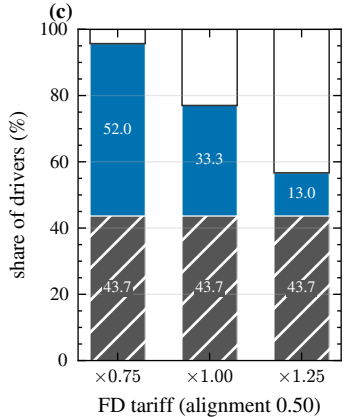
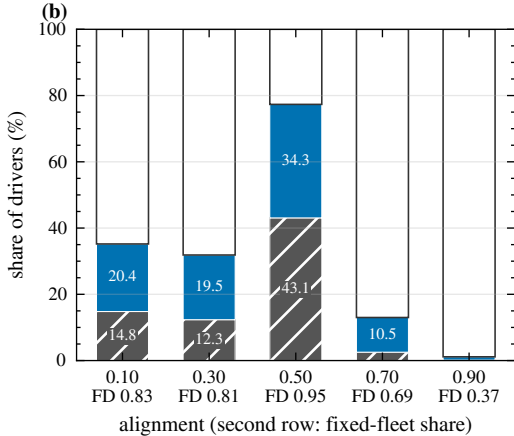
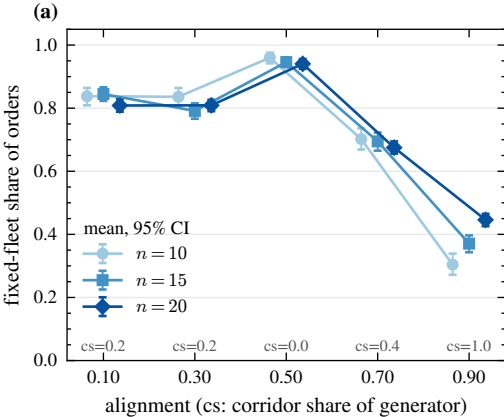




no feasible route ($R_i = \emptyset$) certified FD-dominated ($R_i \neq \emptyset, \mathcal{K}_i^* = \emptyset$) non-empty frontier